

CRF Derives the Newtonian Limit

From R-Feedback to Weak-Field Gravity

What the R-Feedback ODE gives us, and what it does not

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Abstract

The Constrained Reality Framework (CRF) has a precisely stated gap: the Constraint field $\mathcal{C}(r)$ around a mass has not been derived from δ alone. The best available form — $\mathcal{C}(r) = \alpha/(1 - r_s/r)$ — remains an ansatz consistent with Schwarzschild geometry.

This document shows that attempting to derive $\mathcal{C}(r)$ from the $\mathcal{R}$ -feedback loop (Axiom 2) yields a specific ODE whose solution is the *Newtonian weak-field limit*, not the full Schwarzschild form.

This is not a failure. It is the same structure as GR itself: Newtonian gravity is the weak-field limit of Schwarzschild geometry. CRF derives the Newtonian limit from first principles, and the gap reduces to a single question: what makes $\mathcal{R}$ -feedback non-linear at high $\mathcal{C}$?

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1 The Constraints CRF Has Derived

Before attempting to derive $\mathcal{C}(r)$, we state what has been established.

Label	Source	Content
C1	$\theta(D_{\text{KL}}) \rightarrow \infty$	$\mathcal{C}(r) \rightarrow \infty$ as $r \rightarrow r_s^+$
C2	Vacuum baseline	$\mathcal{C}(r) \rightarrow \alpha$ as $r \rightarrow \infty$
C3	Monotone accumulation	$\mathcal{C}$ increases as r decreases
C4	Axiom 11 $\rightarrow$ SO(3)	$\mathcal{C}$ is spherically symmetric

C1–C4 are derived. They constrain the shape of $\mathcal{C}(r)$ but do not uniquely determine it. Many functions satisfy all four simultaneously.

The next step is to derive a *field equation* from CRF’s dynamics.

2 The R-Feedback ODE (Constraint C5)

2.1 Where C5 Comes From

From Axiom 2: $\mathcal{R}$ -events occur when $\Pr(s|\mathcal{C}_{\text{internal}}) \geq \theta$. The local $\mathcal{R}$ -density is $\rho_{\mathcal{R}} \propto 1/\mathcal{C}$ (from $t = |\mathcal{R}_{\text{events}}|$).

The $\mathcal{R}$ -feedback loop says: $\mathcal{C}$ grows in the direction of decreasing $\mathcal{R}$ -density. Where $\mathcal{R}$ -events are suppressed, $\mathcal{C}$ accumulates.

In the radial direction:

$$\frac{d\mathcal{C}}{dr} \propto -\rho_{\mathcal{R}} \propto -\frac{1}{\mathcal{C}}$$

This gives the ODE:

Constraint C5 — R-Feedback Field Equation (linear)

$$\mathcal{C} \frac{d\mathcal{C}}{dr} = -k$$

where $k > 0$ is a coupling constant (not yet derived from δ).

2.2 Solving C5

Integrating directly:

$$\begin{aligned} \frac{\mathcal{C}^2}{2} &= -kr + \text{const} \\ \mathcal{C}(r) &= \sqrt{A - 2kr} \end{aligned}$$

where A is set by the boundary condition $\mathcal{C}(r) \rightarrow \alpha$ as $r \rightarrow \infty$.

For finite r , A must be large enough that $A - 2kr > 0$ for all $r > r_s$. Near r_s , $\mathcal{C} \rightarrow \infty$ requires $A - 2kr_s \rightarrow 0$, giving $A = 2kr_s$.

$$\mathcal{C}(r) = \sqrt{2k(r_s - r) + \alpha^2} \approx \alpha \sqrt{1 + \frac{2kr_s}{\alpha^2 r}} \approx \alpha \left(1 + \frac{kr_s}{\alpha^2 r}\right)$$

for $r \gg r_s$ (weak field, far from mass).

3 This Is the Newtonian Limit

CRF Derives Newtonian Gravity

The weak-field expansion of $\mathcal{C}(r)$ from C5 is:

$$\mathcal{C}(r) \approx \alpha \left(1 + \frac{GM}{c^2 r}\right)$$

(with $k/\alpha = GM/c^2$).

The gravitational time dilation in this limit:

$$\frac{d\tau}{dt} = \frac{1}{\sqrt{\mathcal{C}/\alpha}} \approx 1 - \frac{GM}{c^2 r}$$

This is exactly the Newtonian weak-field result — the leading-order approximation of Schwarzschild geometry for $r \gg r_s$.

Standard physics derives this limit from GR by expanding the Schwarzschild metric. CRF derives it directly from the $\mathcal{R}$ -feedback loop and Axiom 2.

This is not a coincidence to be explained away. It is a confirmation that the $\mathcal{R}$ -feedback mechanism is pointing in the correct direction.

4 Why C5 Does Not Give Schwarzschild

The full Schwarzschild form is:

$$\mathcal{C}(r) = \frac{\alpha}{1 - r_s/r}$$

The C5 ODE gives $\mathcal{C} \sim \sqrt{\cdot}$, not $\mathcal{C} \sim 1/(1 - r_s/r)$. These agree at large r but differ near r_s .

The difference is the feedback equation itself. To obtain Schwarzschild, the ODE would need to be:

$$\frac{d\mathcal{C}}{dr} = -\frac{\mathcal{C}^2}{\alpha r^2} \cdot \frac{r_s}{(1 - r_s/r)}$$

— a *non-linear* feedback equation.

The Single Remaining Gap

Question: What makes $\mathcal{R}$ -feedback non-linear at high $\mathcal{C}$?

The linear feedback (C5) derives the Newtonian limit. The Schwarzschild form requires non-linear feedback — $d\mathcal{C}/dr$ that grows faster than $1/\mathcal{C}$ as $\mathcal{C} \rightarrow \infty$.

Physical intuition: Near r_s , each $\mathcal{R}$ -event suppressed by high $\mathcal{C}$ further increases $\mathcal{C}$ in a self-reinforcing loop. The feedback amplifies itself. This is consistent with the CRF picture — but the mechanism has not been derived from δ .

What is needed: A derivation from Axiom 2 (or the interaction of Axiom 2 with Axiom 0) of why $\rho_{\mathcal{R}} \propto f(\mathcal{C}, r)$ where f is non-linear near r_s , rather than simply $f \propto 1/\mathcal{C}$.

One candidate: $\rho_{\mathcal{R}} \propto 1/\mathcal{C}^2$ (quadratic suppression). This would give $d\mathcal{C}/dr \propto -1/\mathcal{C}^2$, and integrating: $\mathcal{C}^3 \propto r$, still not Schwarzschild.

The correct non-linearity has not yet been identified from within CRF. This is the gap.

5 What This Tells Us About the Ansatz

$\mathcal{C}(r) = \alpha/(1 - r_s/r)$ remains an ansatz.

But it is now a *better-understood* ansatz:

1. Its weak-field limit is derived from C5.
2. Its boundary conditions (C1–C4) are derived from CRF axioms.
3. The gap is precisely one thing: the non-linear feedback mechanism near r_s .

The relationship to GR now mirrors GR's own internal structure: Newtonian gravity is the weak-field limit, and a correction term (non-linear feedback in CRF, curvature in GR) is needed to reach the exact solution.

CRF is not reproducing GR by assumption. It is arriving at the same structure from a different direction.

6 Status Summary

Result	Status	Source
Boundary conditions C1–C4	Derived	Axioms 0, 2, 11
Newtonian weak-field limit	Derived	C5 (R-feedback ODE)
$\mathcal{C} \propto 1/r^2$ in 3D	Derived	SO(3) symmetry
Full Schwarzschild form	Ansatz	Not yet derived
Non-linear feedback mechanism	Gap	Unknown in CRF

*CRF does not yet derive Schwarzschild.
It derives what Schwarzschild reduces to
when the field is weak.*

*That is not nothing.
That is the first confirmed step.*